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MIT Art, Design and Technology University
MIT School of Computing, Pune

Department of Computer Science and Engineering

Assignments- 3&4

Course – Agentic AI

Class - L.Y. (SEM-I), AIA

Name of the Course Coordinator

Dr. Swapnili Karmore

Team Members

A.Y. 2026 - 2027

ASSIGNMENT	
Assignment No: 03	**Unit: Unit III -- Memory, Tools, and Environment Interaction**
Title: Retrieval-Augmented Agent with Long-Term Memory and Tool/Function Calling	
Mapped Course Outcome:	CO3 -- Integrate tools, APIs, and memory modules in intelligent agents. (Apply)
Objective:	To enable students to build an autonomous agent that combines short-term and long-term memory with a vector database, and interacts with external tools/APIs via function calling to complete multi-turn, multi-session tasks.
Problem Statement:	Build a personal-assistant style agent (e.g., a research assistant or coding assistant) using LangChain/LangGraph or AutoGen that maintains short-term conversational memory, stores and retrieves long-term facts using a vector database (FAISS/Chroma), and calls at least two external tools via function calling to complete user requests across multiple sessions.
Implementation Tasks:	1. Set up short-term memory (a conversation buffer) that retains context within a single session. 2. Implement long-term memory using a vector database: embed and store facts/documents, and retrieve relevant context via similarity search. 3. Implement at least two tool-calling functions (e.g., web search, calculator, or code execution) integrated with the agent via function/API calling. 4. Build a feedback-based adaptation loop in which the agent updates its stored memory based on new information or user corrections. 5. Test the agent across 3 multi-turn scenarios that require memory recall from earlier sessions. 6. Log and evaluate retrieval accuracy (relevant memory correctly retrieved) and tool-call success rate. 7. Prepare a 1-page report on the agent's memory-tool architecture, including an interaction-flow diagram.
Tools & Technologies:	Python, LangChain/LangGraph or AutoGen, FAISS/Chroma/Pinecone, an LLM API (OpenAI/Gemini), Jupyter/Google Colab.
Expected Deliverables:	• Working notebook/application implementing the memory + tool-calling agent. • Vector-database indexing and retrieval demonstration. • Multi-turn test log with retrieval-accuracy and tool-call success metrics. • 1-page architecture report with an interaction-flow diagram. • Live demo answering evaluator questions that require memory recall.
Evaluation Criteria	
Criterion	**Marks**
Short-term memory / context retention implementation	2
Long-term memory via vector database -- embedding, storage and retrieval accuracy	3
Tool/function calling integration (minimum 2 tools)	2
Feedback-based adaptation and multi-turn testing	2
Architecture report and live	1

demo/viva	
Total	**10**
Reference / Resources:	• LangChain / LangGraph official documentation • AutoGen tutorials (Microsoft) • FAISS / Chroma vector-database documentation • Unit III syllabus notes -- Memory, Tools and Environment Interaction
Faculty Signature:	**HOD Signature:**

ASSIGNMENT	
Assignment No: 04	**Unit: Unit IV -- Multi-Agent Systems and Collaboration**
Title: Collaborative Multi-Agent Team for Complex Task Completion using CrewAI/AutoGen	
Mapped Course Outcome:	CO4 -- Design collaborative multi-agent systems for complex tasks. (Create)
Objective:	To enable students to design a team of role-based agents that communicate, coordinate, and negotiate to jointly solve a complex task, demonstrating multi-agent communication protocols and coordination strategies.
Problem Statement:	Using CrewAI or AutoGen, design a multi-agent system of at least 3 role-based agents (e.g., Researcher, Writer/Coder, Reviewer/Critic) that collaborate end-to-end to complete a complex task (e.g., researching a topic and producing a technical report, or designing, coding, and reviewing a small software feature). The system must include agent-to-agent communication, a coordination/negotiation mechanism, and at least one human-in-the-loop checkpoint.
Implementation Tasks:	8. Define roles, goals, and available tools for at least 3 agents (e.g., Researcher, Executor/Coder, Reviewer). 9. Implement an agent communication protocol (message passing or a shared task board) between the agents. 10. Implement a coordination mechanism (sequential hand-off, manager/orchestrator agent, or negotiation) to resolve task allocation. 11. Add a human-in-the-loop checkpoint where a human can approve or reject an agent's intermediate output. 12. Run the multi-agent system end-to-end on the chosen task and capture the full interaction/communication log. 13. Introduce a conflict scenario (e.g., two agents producing contradictory outputs) and document how the system resolves it. 14. Prepare a 1-page report analysing coordination effectiveness and comparing the result against a single-agent baseline.
Tools & Technologies:	Python, CrewAI or AutoGen (Microsoft), an LLM API (OpenAI/Gemini), Jupyter/Google Colab.
Expected Deliverables:	• Multi-agent system code with 3 or more clearly defined roles. • Full interaction/communication log for the end-to-end task run. • Conflict-resolution test case and its outcome. • 1-page comparison report: multi-agent system vs. single-agent baseline. • Live demo of the agent team completing the task.
Evaluation Criteria	
Criterion	**Marks**

Role definition and agent setup (3 or more agents)	2
Communication protocol implementation	2
Coordination/negotiation mechanism and conflict resolution	2
Human-in-the-loop checkpoint and end-to-end task completion	2
Comparison report and live demo/viva	2
Total	**10**
Reference / Resources:	• CrewAI official documentation and tutorials • AutoGen tutorials (Microsoft) -- https://microsoft.github.io/autogen/ • Gerhard Weiss, Multi-Agent Systems (Reference Book 4) • Unit IV syllabus notes -- Multi-Agent Systems and Collaboration
Faculty Signature:	**HOD Signature:**

Prepared By

Dr Swapnili Karmore

Note:

In last column L1 is Blooms Taxonomy Level and CO2 Course Outcome. Write Blooms Taxonomy Level and Course Outcome in front of each question in short-hand notation

ent No. 03 -- Unit III -- Memory, Tools, and Environment Interaction

Title: Retrieval-Augmented Agent with Long-Term Memory and Tool/Function Calling

Assessment Criterion	Max Marks	Excellent	Good	Satisfactory	Needs Improvement	Poor
Short-term memory / context retention implementation	**2**	(2 marks) Conversation buffer correctly retains context across multiple turns within a session, seamlessly.	(1.5 marks) Context retained with minor lapses across turns.	(1 mark) Context retention partial or inconsistent.	(0.5 marks) Context retention barely functional.	(0 marks) Short-term memory missing or non-functional.
Long-term memory via vector DB -- embedding, storage & retrieval accuracy	**3**	(3 marks) Vector DB correctly set up; embeddings, storage, and retrieval accurate and well-tested across queries.	(2.5 marks) Vector DB functional with reasonable retrieval accuracy; minor gaps.	(2 marks) Vector DB partially functional; retrieval inconsistent or untested.	(1 mark) Vector DB barely functional; retrieval largely inaccurate.	(0-0.5 marks) Long-term memory/vector DB missing or not attempted.

Tool/function calling integration (min. 2 tools)	**2**	(2 marks) 2+ tools integrated and called correctly via function calling; results used meaningfully by the agent.	(1.5 marks) 2 tools integrated with minor issues in call reliability.	(1 mark) Only 1 tool integrated or calls partially reliable.	(0.5 marks) Tool integration barely functional.	(0 marks) Tool/function calling missing or not attempted.
Feedback-based adaptation & multi-turn testing	**2**	(2 marks) Agent adapts memory correctly based on feedback; all 3 multi-turn scenarios pass with accurate recall.	(1.5 marks) Adaptation mostly works; most multi-turn scenarios pass.	(1 mark) Adaptation partial; some multi-turn scenarios fail recall.	(0.5 marks) Adaptation minimal; recall largely fails.	(0 marks) Feedback adaptation/multi-turn testing missing or not attempted.
Architecture report & live demo/viva	**1**	(1 mark) Clear architecture report with diagram; live demo confidently answers all evaluator questions.	(0.75 marks) Report adequate; demo mostly successful.	(0.5 marks) Report superficial or demo partially successful.	(0.25 marks) Report/demo largely inadequate.	(0 marks) Report and demo missing or unable to demonstrate.
Total	**10**					

Assignment No. 04 -- Unit IV -- Multi-Agent Systems and Collaboration

Title: Collaborative Multi-Agent Team for Complex Task Completion using CrewAI/AutoGen

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Assessment Criterion	Max Marks	Excellent	Good	Satisfactory	Needs Improvement	Poor
Role definition & agent setup (3+ agents)	**2**	(2 marks) 3+ agents with clearly defined, distinct roles, goals and tools; well documented.	(1.5 marks) 3 agents defined with minor overlap/ambiguity in roles.	(1 mark) Fewer than 3 agents or roles poorly defined.	(0.5 marks) Agent roles barely distinguishable.	(0 marks) Agent role definition missing or not attempted.
Communication protocol implementation	**2**	(2 marks) Robust message-passing/shared task-board protocol; agents exchange information reliably.	(1.5 marks) Protocol functional with minor communication gaps.	(1 mark) Protocol partially functional; some messages lost/ignored.	(0.5 marks) Communication barely functional.	(0 marks) Communication protocol missing or not attempted.
Coordination/negotiation mechanism & conflict resolution	**2**	(2 marks) Coordination mechanism clearly implemented; conflict scenario resolved sensibly and documented.	(1.5 marks) Coordination functional; conflict resolution mostly reasonable.	(1 mark) Coordination present but conflict handling weak/undocumented.	(0.5 marks) Coordination minimal; conflict unresolved.	(0 marks) Coordination mechanism missing or not attempted.
Human-in-the-loop checkpoint & end-to-end task completion	**2**	(2 marks) Checkpoint correctly implemented; task completed end-to-end with human approval integrated smoothly.	(1.5 marks) Checkpoint present; task completed with minor issues.	(1 mark) Checkpoint present but task only partially completed.	(0.5 marks) Checkpoint barely functional; task mostly incomplete.	(0 marks) Human-in-the-loop checkpoint or task completion missing.
Comparison report & live demo/viva	**2**	(2 marks) Clear multi-agent vs single-agent comparison; live demo confident and accurate.	(1.5 marks) Report adequate; demo runs with minor issues.	(1 mark) Report superficial or demo partially functional.	(0.5 marks) Report missing key content; demo largely non-functional.	(0 marks) Report/demo missing entirely.
Total	**10**					